

Product Requirements Document (PRD)

Product Name

AI Mock Test Generator (Working Title)

Document Purpose

This PRD defines the product vision, user problems, scope, functional and non-functional requirements for building an AI-powered, universal CBT mock test platform for Indian competitive exams.

1. Problem Statement

In India, the majority of competitive examinations (government jobs, engineering, medical, and state-level entrance exams) are conducted in Computer Based Test (CBT) mode. Millions of aspirants attempt these exams every year.

However, aspirants face the following problems: - Insufficient number of quality CBT mock tests - High cost of exam-specific mock test series - Fragmented platforms for different exams - Lack of realistic exam interface and timing pressure - Poor performance analytics and feedback

There is no single, affordable platform that provides unlimited, exam-standard CBT mock tests across multiple Indian exams.

2. Product Vision

To build a **universal, AI-powered CBT mock test platform** that enables aspirants to generate and attempt unlimited, realistic mock tests for Indian competitive exams, helping them improve speed, accuracy, and exam confidence.

3. Goals & Success Metrics

Primary Goals

- Provide realistic CBT exam simulation
- Enable unlimited mock test generation
- Support multiple exams using a single engine
- Improve aspirant performance through analytics

Success Metrics (MVP)

- Users can complete a full-length CBT mock
 - Average test completion rate > 70%
 - Positive feedback on exam realism
 - Repeat usage by aspirants
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4. Target Users

Primary Users

- Government job aspirants (SSC, Banking, Railways)
- Engineering entrance aspirants (JEE, GATE)
- Medical entrance aspirants (NEET)
- State-level exam aspirants (EAMCET, CETs)

Secondary Users

- College students preparing for multiple exams
 - Coaching institutes (future B2B)
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5. In-Scope vs Out-of-Scope (MVP)

In-Scope (MVP)

- CBT-based mock tests (MCQ)
- 1–2 supported exams initially
- Exam configuration-driven system
- Static + AI-generated questions
- Auto-evaluation
- Basic analytics

Out-of-Scope (MVP)

- Descriptive answer evaluation
 - Interview preparation
 - Mobile applications
 - Full-scale AI personalization
 - Offline mode
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6. Core User Flow

1. User signs up / logs in

2. Selects target exam
 3. Chooses mock type (full-length / section-wise)
 4. Starts CBT mock test
 5. Attempts questions within timed environment
 6. Submits test
 7. Views score and analytics
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7. Key Features & Requirements

7.1 Exam Configuration Engine

- Exam patterns defined via configuration (JSON/YAML)
- Parameters include:
 - Sections
 - Number of questions
 - Time limits
 - Negative marking rules

Requirement: Ability to add new exams without code changes.

7.2 CBT Test Interface

Functional Requirements: - Question display with options - Timer (overall + section-wise) - Question navigation panel - Mark for review - Auto-submit on time expiry

Non-Functional Requirements: - Responsive UI - Low latency - Accurate timer handling

7.3 Question Management System

Sources: - Seed question bank (PYQs) - AI-generated questions

Requirements: - Topic-wise tagging - Difficulty tagging - Randomized question selection - No repetition within a test

AI Constraints: - AI generates language only - Answers validated programmatically

7.4 AI Question Generation

Functional Requirements: - Generate questions based on: - Exam - Topic - Difficulty - Follow predefined question blueprints

Quality Control: - Programmatic answer verification - Similarity check against PYQs - Difficulty calibration

7.5 Evaluation & Analytics

Basic Analytics (MVP): - Total score - Correct vs incorrect - Time spent per section - Accuracy percentage

Future Analytics (Post-MVP): - Topic-wise weakness detection - Difficulty-adjusted scoring - Personalized recommendations

8. Non-Functional Requirements

- Scalability to support thousands of concurrent users
 - Secure storage of user data
 - High availability during peak hours
 - Clear separation of AI logic and exam logic
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9. Assumptions & Constraints

Assumptions: - Exam patterns are stable and known - Users have basic internet access - CBT familiarity exists among aspirants

Constraints: - Limited initial dataset - Cost constraints on AI API usage - Small development team

10. Risks & Mitigation

Risk	Mitigation
AI generates wrong questions	Validation layer
Difficulty mismatch	Analytics-based calibration
Copying PYQs	Similarity detection
Low user trust	Realistic CBT UI

11. Phased Roadmap

Phase 1 – MVP

- Single exam support
- CBT interface
- Static question bank
- Basic analytics

Phase 2

- AI-generated questions
- Multiple exams
- Topic-wise mocks

Phase 3

- Advanced analytics
 - Personalization
 - B2B offerings
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12. Conclusion

The AI Mock Test Generator aims to solve a critical gap in India's competitive exam preparation ecosystem by offering an affordable, scalable, and exam-realistic CBT mock test platform.

The product focuses on quality, standardization, and accessibility, making it suitable for millions of aspirants across domains.